INCH-POUND

MIL-PRF-49471/1A(CR)
30 November 2000
Superseding
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2 June 1995

PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, HIGH PERFORMANCE, BA-X599/U

This specification sheet is approved for use within the Communications Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification and MIL-PRF-49471(CR).

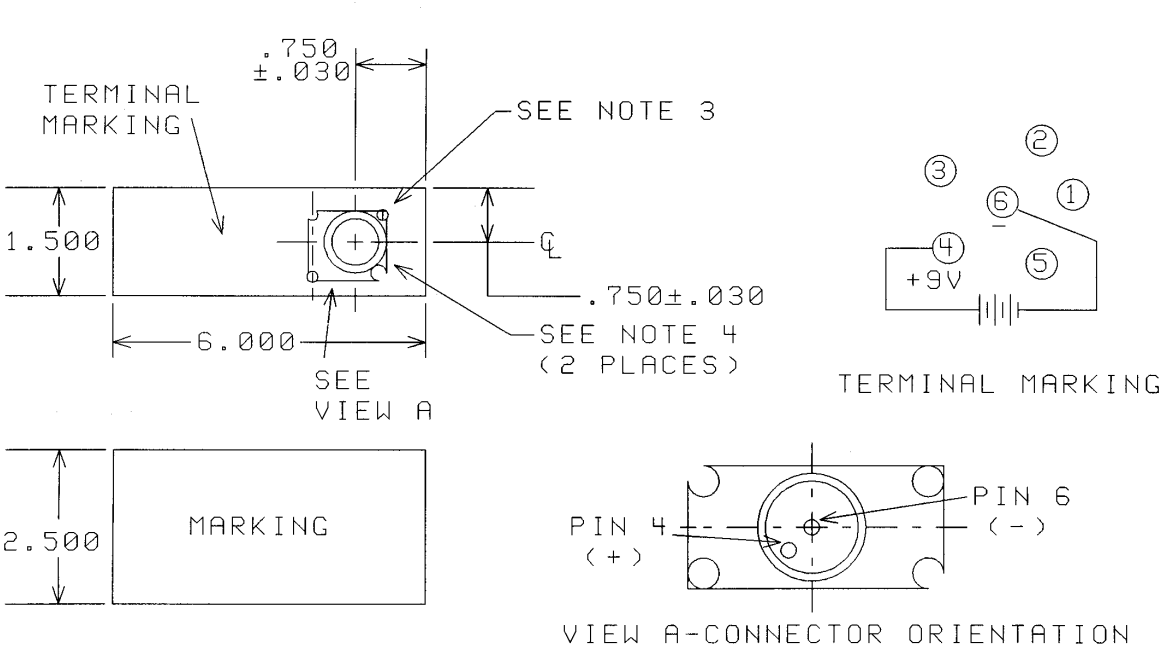

Figure 1. Battery dimensions and terminal markings.

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NOTES:

1. All dimensions are in inches.

2. Tolerances, unless otherwise specified, are -.063, +.000.

3. The battery terminal shall consist of a miniature circular battery connector, Dwg SC-C-179492, except as noted herein.

4. Connector socket body configuration shall accept two (2) each 0.156 inch minimum diameter guideposts as shown. Depth of cavity shall be 0.657 inch minimum.

5. The socket shall be supported and mounted so that the top surface of the socket shall not protrude above the adjacent outside surface of the jacket and shall not be more than 1/16 inch below adjacent outside surface of the jacket when used initially, during, and after subsequent insertions of the mating plug during contractual testing.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Maximum voltage: 9.90 volts.

Cut-off voltage for capacity: 6.00 volts

Terminals: Six (6) hole socket type with no obstructions of any holes (see notes 3 and 5).

Weight (maximum): 1.5 pounds (680 grams).

Charge protection: Applicable.

Battery Short Circuit Test: Applicable.

State of Charge Indicator: Not applicable

High temperature protection: Applicable.

Complete discharge device: Applicable.

Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

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Capacity Test	Minimum Capacity Requirements in Hours to 6.00 Volts
I	3.2
L	1.7
H	6.2
HT	6.0
LT	1.6
IT	3.0
LR1	47.0

Initial voltage delay: During the I, H, HT, and IT discharge tests, initial voltages below 6.0 volts cannot exceed a 5.0 second duration. During the L and LT discharge tests initial voltages below 5.70 volts cannot exceed a 60.0 second duration.

Connector: The connector configuration and operating temperature range requirements shall be certified in the First Article Test Report. The connector shall be in accordance with SC-C-179492 with the following exceptions: (1) Sheet two of SC-C-179492 does not apply. (2) Mounting screws are optional. (3) Operating temperature range for the connector shall be from a low of-55°C to a high of 125°C. (4) The connector test described in paragraph 4.7.15 shall be performed 10 times with SM-D-687888. (5) Each of the metallic sockets in the connector (top view) shall conform to Figure 2. Each socket shall be cylindrical in shape and may have a single gap with a maximum opening of 30 degrees. The socket shall accept a number 16 pin.

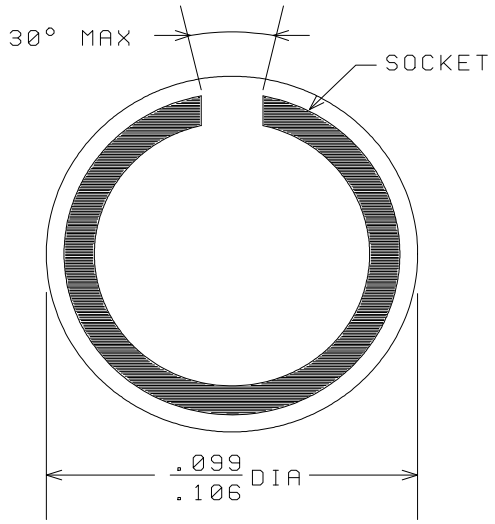

Figure 2. Socket (Top View)

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METHODS OF EXAMINATION AND TESTS:

Capacity tests:

(1) Storage:

Details on storage conditions for all capacity tests are specified in the basic specification.

(2) Discharge:

Cells (forced discharge): Two cells shall be discharged at 2.0 amperes at 70±5°F to 2.0 volts Each discharged cell shall then be connected in series to two fresh cells. Each string shall be force discharged at 2.0 amperes at 70±5°F for 3.2 hours.

Batteries: For all capacity tests, except the LR1, H and HT, the batteries shall be discharged at 2.0 amperes constant current to cut-off volts (zero volts for the I test). For the H and HT capacity tests the batteries shall be discharged through 2.0 amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches cut-off volts. For the LR1 test, batteries shall be discharged at 135 milliamperes to 6.5 volts, then at 1.7 watts to 1.0 volt.

Forced discharge: After discharge to zero volts on the I capacity test, the batteries shall be force discharged at 2.0 amperes for a minimum of 5 minutes.

Overload test: Batteries shall be discharged at 4.0 amperes.

Abuse test predischarge: Designated samples shall be discharged at 2.0 amperes for 1.6 hours.

Abuse test, pulse discharge: Batteries shall be discharged at 2.5 amperes for 1 minute followed by 1.8 amperes for 4 minutes, cycled continuously to cut-off voltage.

Closed-circuit voltage test (Battery & cell string tests): Load resistance of 3 ohms shall be used. Voltage shall be above 6.00 volts within 10.0 seconds.
Cell closed-circuit voltage test: Load resistance of 1 ohm shall be used. Voltage shall be above 2.0 volts in 5 seconds.

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Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army-CR

Preparing Activity:
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